WEEK 4 REPORT - POWER BI EDA

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INTRODUCTION:

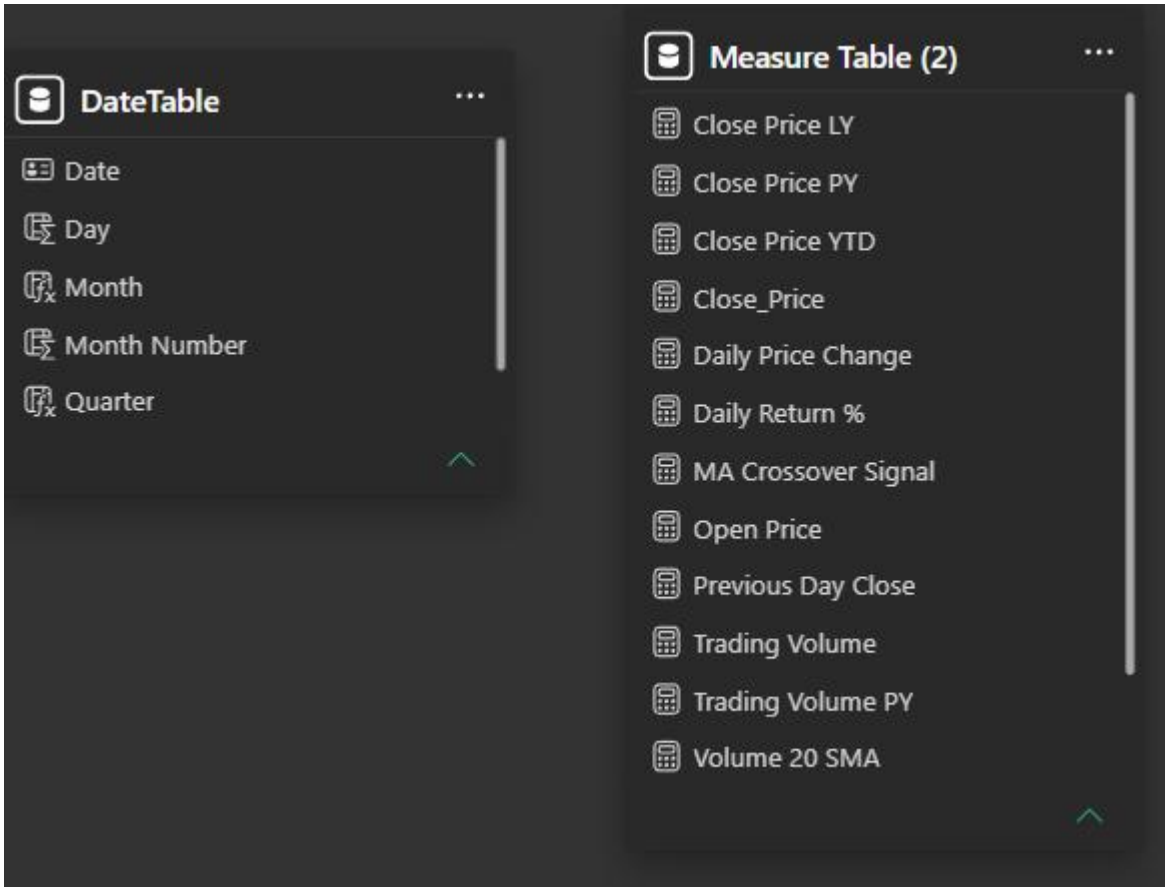
This week, the Shopify stock market data was analyzed using Power BI. The data was prepared for analysis, and DAX measures were created to study stock prices and trading volume. Moving averages such as 20-Day SMA and 50-Day SMA were used to understand the stock trend. Interactive charts, slicers, and a tooltip were also added to make the dashboard easier to use and understand.

DATA PREPARATION:

The Shopify stock dataset was imported into Power BI and prepared for analysis. The date field was used to organize the stock data by different time periods. The data was also arranged to analyze stock prices and trading volume using Power BI visuals and DAX measures.

DAX MEASURES:

DAX measures were created to calculate the 20-Day SMA, 50-Day SMA, and Moving Average Crossover Signal. These measures were used to analyze moving-average trends in the stock data.



TRADING VOLUME ANALYSIS:

The trading volume was analyzed to understand how the stock activity changed over time. A 20-Day Simple Moving Average (SMA) and 50-Day Simple Moving Average was used to understand the overall volume trend. The chart shows the changes in trading volume and helps to identify periods of high and low trading activity. This makes it easier to understand the changes in market activity during different time periods.

```
Trading Volume =
SUM('SHOP_2015-05-21_2025-03-16'[Volume])
```

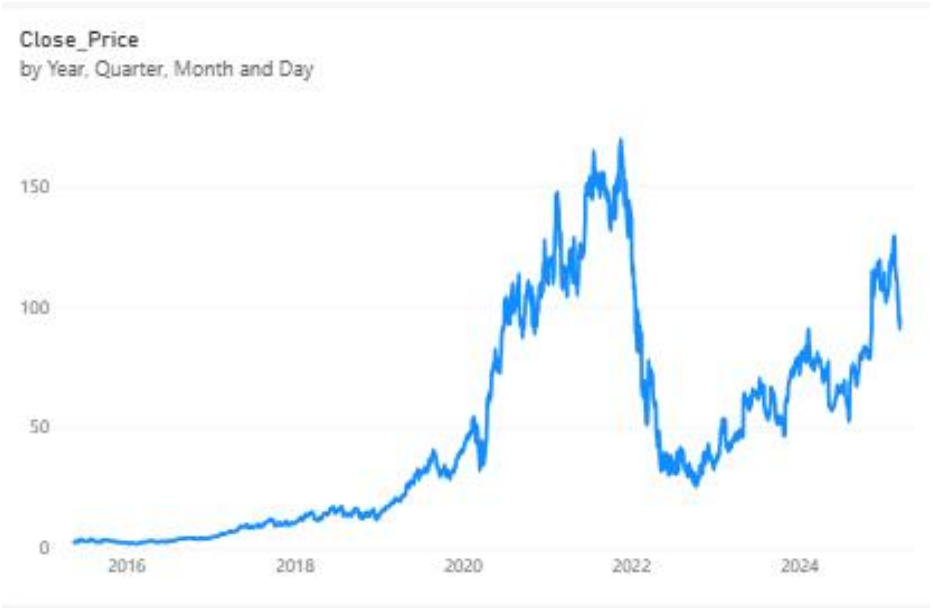
```
Trading Volume PY =
CALCULATE(
    [Trading Volume],
    SAMEPERIODLASTYEAR('DateTable'[Date])
)
```

```
Volume 20 SMA =
VAR CurrentDate = MAX('DateTable'[Date])
VAR PeriodDays = 20
VAR DateInPeriod =
    DATESINPERIOD(
        'DateTable'[Date],
        CurrentDate,
        -PeriodDays,
        DAY)
RETURN
    CALCULATE(
        AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
        DateInPeriod
    )
```

```
Volume 50 SMA =
VAR CurrentDate = MAX('DateTable'[Date])
VAR PeriodDays = 50
VAR DateInPeriod =
    DATESINPERIOD(
        'DateTable'[Date],
        CurrentDate,
        -PeriodDays,
        DAY
    )
RETURN
    CALCULATE(
        AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
        DateInPeriod
    )
```

PRICE TREND ANALYSIS:

The Close Price was analyzed to understand how the Shopify stock price changed over time. The price chart shows the movement of the stock from 2015 to 2025, including the rise and fall in prices during different periods. The Daily Return % was also used to observe the daily changes in the stock price. This helps to understand the overall price movement and changes in stock performance.



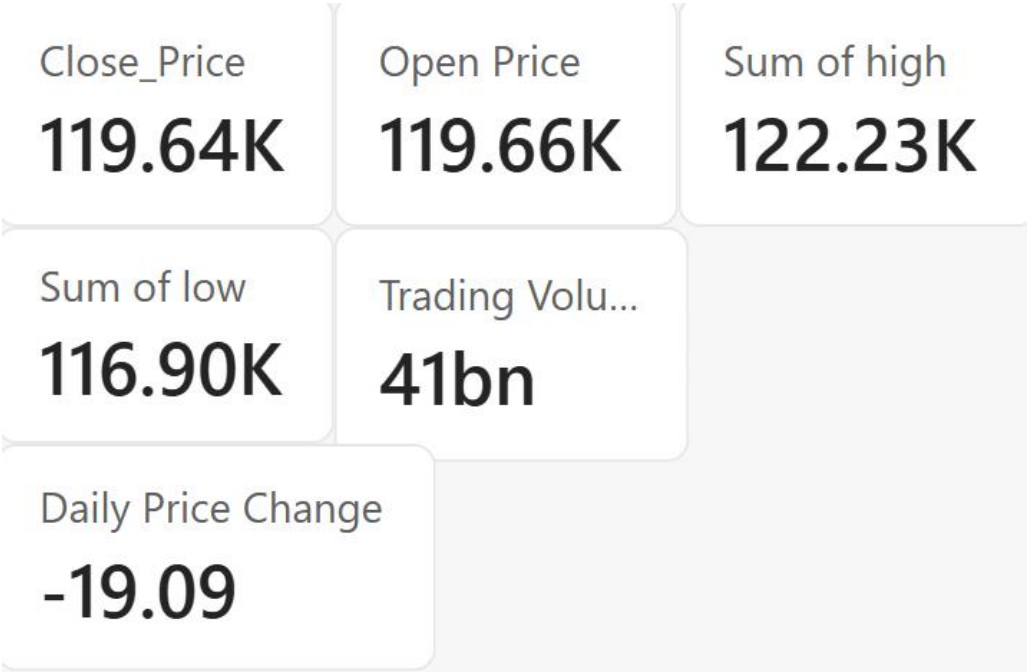
SLICERS AND DYNAMIC SELECTION:

A Month slicer was added to the dashboard to filter the stock data based on different months. This allows the user to view the required time period easily. The slicer makes the dashboard more interactive and helps users focus on the stock price and trading volume data they want to analyze.



CUSTOM TOOLTIP:

A custom Stock_Tooltip page was created to show important stock details when the user moves the mouse over the chart. It displays information such as Open Price, High Price, Low Price, Close Price, Trading Volume, and Daily Change %. This makes it easier to view detailed stock information without adding too much data to the main dashboard.



CONCLUSION:

The Week 4 analysis improved the Shopify stock market dashboard by using DAX measures and moving averages to study the stock data. Trading volume and price trends were analyzed to understand changes over time. A Month slicer and interactive dashboard features were also added to make the analysis easier. The custom tooltip provided additional stock details such as price and volume when viewing the charts.

Overall, these features made the Power BI dashboard more interactive and easier to understand.